

Zhenpeng Li

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EDUCATION

Department of Mathematics, University of Wisconsin-Madison

Ph.D. Candidate

Madison, U.S.

In Progress

School of Mathematical Science, Peking University

Bachelor of Science in Mathematics Science

Beijing, China

September 2020 – June 2024

- **GPA: 3.80/4.0, Major GPA: 3.91/4.0**
- **Selected Courses:**

Fiber Bundles and Characteristic Classes 100.	Modular Forms and Number Theory 95.
Algebraic Geometry I 92.	Geometry 95.5.
Abstract Algebra (Honor): I 98; II 94.	Advanced Algebra: I 97; II 95.
Mathematical Analysis (Honor): I 98; II 94.	Functions of Real Variables 99.
Functional Analysis 96.5.	Riemann Surfaces 92.
Differentiable Manifolds 90.	Functions with Complex Variables (Honor) 90.
Probability Theory 97.	Homological Algebra 90.
Chromatic Homotopy Theory 97.	Homology Theory P.
Étale Cohomology Theory 94.	Group Theory P.
Topics in Algebraic Topology 97.	Game Theory 98.
Computation (A) 92.	

ACADEMIC INTERESTS

Algebraic topology: (chromatic, motivic, equivariant) homotopy theory, algebraic K -theory

Related interests: algebraic number theory, algebraic geometry, (geometric) representation theory

PUBLICATION & PREPRINTS

Preprints:

1. *The spectrum of strict units of topological modular forms*, with Kiran Luecke, available at [arXiv:2607.17124](https://arxiv.org/abs/2607.17124).
2. *Normed equivariant ring spectra and $\mathbb{A}_n$ -modules*, with Kailin Pan, in preparation.

HONORS & AWARDS

Winning Prize (top 0.03% worldwide), in 2024 Alibaba Global Mathematics Competition, for algebra and number theory 2024

First Prize (rank 1), Undergraduate Research Competition 2024

First Prize (rank 9), Excellent Bachelor Thesis Competition 2024

Winning Prize (rank 10 in China), 14'th session of S.T. Yau's College Student Mathematical Contests, for algebra and number theory 2023

Merit Student of Peking University 2023

Encouragement Scholarship from Sigma Square Tech 2023

Encouragement Scholarship for pure mathematics from Hope Online 2023

First Prize, 13'th session of National College Student Mathematics Contest 2022

Merit Student of Peking University 2021

Third-class Scholarship of Peking University 2021

RELEVANT SEMINAR TALKS

Balmer spectra of Sp_A , Hausmann-Meier, “Invariant prime ideals in equivariant Lazard rings”	Autumn 2024
Pro-étale cohomology of rigid analytic spaces, Sec. 5, BSSW, “On the rationalization of the $K(n)$ -sphere”	Autumn 2024
LMMT’s paper “Purity in chromatically localized algebraic K -theory”	Summer 2024
Breuil-Kisin twist and prismatic log, Sec. 2, Bhatt-Lurie, “Absolute prismatic cohomology”	Summer 2024
CMNN’s paper “Descent and vanishing in chromatic algebraic K -theory via group action”	Summer 2024
BCSY’s paper “The chromatic Fourier transform”	Summer 2024
BHLS’s “ K -theoretic counterexamples to Ravenel’s telescope conjecture” (video available) (notes available)	Summer 2024
Perverse sheaves, Ch. 4, Kiehl-Weissauer, “Weil Conjectures, Perverse Sheaves and l -adic Fourier Transform”	Summer 2024
Efimov K -theory, Sec. 1, 2, and 4, Efimov, “ K -theory and localizing invariants of large categories”	Spring 2024
Carmeli-Schlank-Yanovski’s paper “Chromatic cyclotomic extensions”	Winter 2023
Carmeli-Schlank-Yanovski’s paper “Ambidexterity and height”	Winter 2023
Yonatan Harpaz’s paper “Ambidexterity and the universality of finite spans”	Winter 2023
Sec. 4-6 of BSY’s paper “The chromatic Nullstellensatz”	Winter 2023
Chromatic homotopy theory, Lurie, “Lecture notes on chromatic homotopy theory”	Autumn 2023
Deligne-Mumford stacks and the Keel-Mori theorem, Ch. 4, Alper, “Stacks and moduli”	Autumn 2023
Algebraic K -theory on schemes, Ch. 2-3, Sriniva, “Algebraic K -theory”	Autumn 2023
Sphere Packing Problem, Viazovska, “The Sphere Packing Problem in Dimension 8” (notes available)	Summer 2023
Higher algebra, Lurie, “Kerodon”, “Higher topos theory”, and “Higher algebra”	Spring 2023
Abelian Varieties, Ch. 3, Mumford, “Abelian Varieties”	Spring 2023
Linnik’s conjecture, Kuznecov, “Conjecture for cusp forms of weight zero” (notes available)	Winter 2022
Commutative Algebra, Ch. 9-10, Atiyah-Macdonald	Spring 2022
Algebraic Topology, Sec. 14, 18, Bott-Tu, “Differential Forms in Algebraic Topology”	Spring 2022

TEACHING EXPERIENCE

Teaching assistant on Math 212, Calculus	Spring 2026
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